



AI IN ACCOUNTING (AICA) - LEVEL 2

DESI FINTAX ASG GSTN VERIFIER

An operational Windows application for client-wise GSTIN verification, return-filing analysis and evidence reporting

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Brand	Desi Fintax
Project type	AI-assisted operational application
Deployment	Windows 10/11 installer and standalone executable
Version	1.4.0

Prepared for submission under the AICA Level 2 Certificate Course of ICAI

1. Executive Summary

Project in one sentence: A practical desktop application that converts repetitive supplier-GSTIN checking into a controlled client-wise workflow, while retaining manual CAPTCHA compliance and producing Excel and PDF evidence.

In professional GST work, supplier invoice files often contain the same GSTIN across many invoices. Manually checking every row would repeat the same portal search, waste time and make evidence collection inconsistent. The project began with a simple operational idea: import invoice data, identify each unique GSTIN, verify it only once, and connect the result back to all related invoice values.

The completed Desi Fintax ASG GSTN Verifier is a Windows desktop application for Chartered Accountants, auditors, accountants and tax teams. It maintains separate clients and projects, imports Excel and GSTR-2A/2B data, validates and deduplicates GSTINs, automates permitted portal navigation, captures GSTR-1/IFF and GSTR-3B filing information for the selected financial year, saves evidence PDFs, and exports structured reports.

Artificial intelligence was used as a development partner for requirement analysis, prompt-driven prototyping, debugging, workflow redesign, testing and documentation. The application itself does not use AI to bypass CAPTCHA, impersonate users or access undocumented GST Portal services.

2. Background and Starting Story

The starting problem came from day-to-day CA-office work. A GSTR-2A/2B or invoice workbook may contain hundreds or thousands of invoice rows, but the number of unique suppliers is much smaller. The same supplier GSTIN may appear repeatedly across months and files. The practical requirement was therefore not to verify every invoice row separately, but to verify each unique supplier once and aggregate all connected invoice amounts GSTIN-wise.

The manual process also required opening the GST Portal, entering a GSTIN, completing CAPTCHA when demanded by the portal, opening the filing table, selecting the required financial year, reviewing return history, saving evidence and then repeating the process. An interruption could force the user to start again. The project was designed to reduce this repetition while keeping the portal-controlled CAPTCHA step manual.

3. Problem Statement

- Repeated GSTINs in invoice data lead to duplicate verification effort.
- Manual GST Portal searches are slow and prone to missed financial-year selections or incomplete evidence.
- Client and project records must remain isolated; combining unrelated datasets can produce misleading reports.
- Monthly and quarterly filing patterns are not always directly labelled in the filing table and may change during a year.
- Interrupted browser sessions, CAPTCHA errors and portal downtime need clear recovery options instead of an indefinitely pending queue.

- The final output must be useful in scrutiny, audit and advisory work, while avoiding any incorrect conclusion that filing GSTR-3B proves invoice-wise tax payment or ITC eligibility.

4. Project Objectives

- Create a simple, branded and client-focused Windows application.
- Import invoice and supplier data from common Excel, GSTR-2A/2B and PDF sources.
- Normalise, checksum-validate and deduplicate GSTINs before portal verification.
- Aggregate invoice counts and taxable, tax and invoice values supplier-wise.
- Capture registration information and filing history for the selected financial year.
- Infer monthly or quarterly filing from the sequence of filed tax periods, including changes within a year.
- Generate evidence PDFs and professional Excel/PDF reports.
- Allow pause, resume, retry and recovery after interruptions.
- Provide an installer and executable that work without a separate Python installation.

5. Operational Workflow

1. Create or select a client project and choose the required financial year or month.
2. Import invoice data, a GSTR-2A/2B workbook, a supplier list, or manually entered GSTINs.
3. Validate GSTIN structure and checksum; show genuine invalid entries with their source file, sheet and row.
4. Remove duplicate GSTINs from the verification queue while preserving invoice-level records.
5. Aggregate invoice values GSTIN-wise and link them to the selected client, project and import period.
6. Start portal-assisted verification. The application enters the GSTIN and navigates the workflow; the user completes CAPTCHA whenever the portal requires it.
7. Select and capture the requested financial year's GSTR-1/IFF and GSTR-3B filing rows.
8. Infer filing frequency, calculate general due dates and delay days, and save the result with timestamped evidence.
9. Continue to the next pending GSTIN and stop cleanly when the queue is complete.
10. Export client-wise, period-wise and supplier-wise Excel/PDF reports.

6. Main Functions Delivered

Function	Operational result
Client/project isolation	Dashboard, imports, verification queue, results and reports are filtered by the active project.
GSTIN validation	Normalisation, structural checks and checksum validation before portal processing.
Deduplication	One verification queue item per project/GSTIN, irrespective of repeated invoice rows.
Invoice aggregation	Invoice count and taxable value, IGST, CGST, SGST, cess, GST and invoice value totals.
Period-aware imports	Annual or monthly GSTR-2A/2B context with consolidated and period-wise reporting.

Function	Operational result
Portal assistance	Automatic GSTIN entry and navigation with manual CAPTCHA and visible-browser fallback.
Filing analysis	GSTR-1/IFF and GSTR-3B rows for the selected FY, frequency inference, due date and delay.
Evidence	Short GSTIN/FY PDF filenames, timestamps and evidence indexing.
Recovery	Pause, resume, retry, skip and queue reset without deleting completed verifications.
Deployment	PyInstaller executable and Inno Setup installer for Windows 10/11.

7. Use of AI and Prompt Engineering

The project demonstrates AI-assisted software development rather than merely generating a written report. A detailed master prompt defined the business problem, controls, technology and deliverables. Subsequent prompts were used as structured feedback after actual testing. Each prompt identified an operational failure or missing requirement, and the application was revised and tested again.

- Requirement engineering: translating GST-office work into modules, data fields, reports and safeguards.
- Iterative debugging: converting portal errors, queue interruptions and incorrect FY selection into reproducible fixes.
- Domain logic: defining deduplication, invoice aggregation, FY mapping, monthly/quarterly inference and due-date presentation.
- User-experience improvement: simplifying the dashboard around the active client/project and showing clear recovery messages.
- Quality assurance: generating automated tests for core calculations, project isolation, portal mapping and queue completion.
- Documentation and deployment: preparing user instructions, release notes, installer configuration and the capstone package.

Prompt-file scope: The submitted prompt files stop at completion of the operational v1.4.0 application. Later prompts concerning YouTube promotion, social-media replies, Drive distribution and unrelated marketing have been intentionally excluded.

8. Technology and Frameworks

Framework/library	Purpose in the project
Python 3.12/3.13	Core application language and business logic.
PySide6 / Qt	Native Windows desktop interface, dialogs, tables and workflow controls.
SQLite	Local relational database for projects, invoices, suppliers, verifications, filing rows and audit history.
pandas and openpyxl	Excel/GSTR workbook reading, aggregation and professionally formatted exports.

Framework/library	Purpose in the project
Playwright with Microsoft Edge	Permitted browser-assisted GST Portal navigation and evidence capture.
PyMuPDF and ReportLab	PDF text extraction, report creation and evidence-related processing.
cryptography	Protection of locally stored optional credentials and secrets.
pytesseract (optional)	OCR fallback when a supplied PDF contains scanned images instead of searchable text.
PyInstaller	One-folder Windows executable build.
Inno Setup	Windows installer, shortcuts, upgrade handling and uninstall configuration.
Python unittest	Automated regression testing of critical workflows.

9. Data and Filing Logic

9.1 GSTIN and invoice handling

GSTINs are cleaned, converted to uppercase and validated before they are added to the portal queue. The supplier record is unique within the active project, while every original invoice remains available separately. This permits one portal verification to support many invoice rows without losing reconciliation detail.

9.2 Financial-year selection

The project financial year is mapped to the GST Portal's displayed year, for example FY 2025-26 to 2025-2026. A verification is not treated as complete if the requested FY or its filing rows are absent. Supported choices extend from FY 2017-18 through the ongoing financial year.

9.3 Monthly and quarterly inference

Where the portal table does not directly label frequency, the application evaluates the spacing between displayed tax periods. Consecutive monthly periods indicate monthly filing; quarter-end periods separated by about three months indicate quarterly filing. Mixed patterns are retained period-wise instead of forcing one classification for the entire year.

9.4 Due dates and delays

The application presents clearly labelled general statutory due dates for GSTR-1/IFF and GSTR-3B and calculates the difference between the due date and actual filing date. Because due dates can be extended by notification and can depend on taxpayer facts, the reports expressly require verification of applicable extensions before legal reliance.

10. Security, Ethics and Compliance

- CAPTCHA is never bypassed or automatically solved. It remains a manual user action controlled by the GST Portal.

- The application does not reverse-engineer undocumented GST Portal APIs or treat browser tokens as an unofficial bulk API.
- Data is stored locally by default. No client database or saved credential is included in the capstone submission.
- Optional credentials are encrypted and associated with the relevant client/project; passwords are masked in the interface.
- The tool is independently developed and is not represented as an official GSTN or Government application.
- Return filing is not reported as independent proof of invoice-wise tax payment, supplier tax discharge or recipient ITC eligibility.
- Public GSTINs in the showcase workbook are paired with fictional invoice numbers and values solely for demonstration.

11. Testing and Validation

The final v1.4.0 source was tested on 16 August 2026 using Python's unittest framework. All 19 automated tests passed.

Test group	Verified behaviour
GSTIN core	Normalisation, checksum validation, extraction, bulk deduplication and 100-invoice aggregation.
Persistence/reporting	Cross-import deduplication, history, exports, backup and strict project isolation.
Portal mapping	Official route, FY label matching, public response mapping and GST Portal error explanation.
Filing capture	Full month periods, filing-frequency inference and requested-year handling.
Evidence	PDF screenshot fallback when direct print is unavailable.
Recovery	Legacy migration, reset protection, next-pending selection and clean end-of-queue behaviour.

Result: 19 tests run; 19 passed; 0 failures. The installer and standalone executable were prepared separately for Windows deployment.

12. Benefits to Professional Work

- Reduces repetitive supplier searches by verifying each project GSTIN once.
- Creates a consistent audit trail connecting imported invoices, supplier verification and evidence files.
- Helps identify cancelled, suspended, delayed or non-filing suppliers requiring professional review.
- Allows monthly analysis and annual consolidation without merging unrelated client projects.
- Improves continuity because interrupted queues can resume without losing completed work.
- Produces Excel and PDF outputs suitable for internal review, client communication, scrutiny preparation and working-paper support.
- Demonstrates how domain expertise and AI-assisted development can convert a recurring office problem into an operational tool.

13. Limitations and Professional Caution

The GST Portal controls availability, CAPTCHA, login/MFA requirements, page layout and displayed information. Portal changes can require software maintenance. Live production API access, if later required, must be obtained through an authorised GSP/API provider.

The tool supports professional review; it does not replace statutory interpretation or examination of invoices, books, e-invoices, e-way bills, payment evidence, supplier communications and applicable notifications. Filing status alone cannot determine invoice-level tax payment or ITC entitlement.

14. Future Scope

- Optional authorised GSP/API integration for faster production verification.
- Digitally signed releases and automatic application updates.
- Additional exception analytics and litigation-ready reconciliation schedules.
- Role-based access and controlled multi-user office deployment.
- Expanded import mappings for additional GST reports and accounting-system formats.

15. Conclusion

The Desi Fintax ASG GSTN Verifier addresses a real and recurring GST-office problem: repeated supplier verification across large invoice datasets. It combines GST domain understanding, structured prompting, Python development, portal-assisted automation, evidence preservation and professional reporting in one operational Windows application. The result is a practical demonstration of AI-assisted problem solving for Chartered Accountants, with appropriate safeguards around CAPTCHA, credentials, portal limitations and legal conclusions.

16. Key Technical References

- Python documentation - <https://docs.python.org/3/>
- Qt for Python / PySide6 - <https://doc.qt.io/qtforpython-6/>
- SQLite documentation - <https://www.sqlite.org/docs.html>
- pandas documentation - <https://pandas.pydata.org/docs/>
- openpyxl documentation - <https://openpyxl.readthedocs.io/>
- Playwright for Python - <https://playwright.dev/python/>
- PyMuPDF documentation - <https://pymupdf.readthedocs.io/>
- ReportLab documentation - <https://docs.reportlab.com/>
- PyInstaller documentation - <https://pyinstaller.org/>
- Inno Setup documentation - <https://jrsoftware.org/ishelp/>

Appendix A - Submission Contents

- Project Summary: DOCX and PDF copies of this document.

- Prompt Files: the original master development prompt and curated, relevant refinement prompts only.
- Example Files: public-GSTIN showcase workbook, import template and interface images.
- Executable: Desi Fintax ASG GSTN Verifier v1.4.0 Windows installer.
- Supporting Documents: user manual, README, release notes, requirements and automated test results.
- Source Code: a clean source archive excluding databases, credentials, build caches and user evidence.